CorpusGuard

AI Security Assessment Report

Target System	Test RAG
Assessment Date	July 21, 2026 — 22:29 UTC
Framework Version	CorpusGuard 0.1.0
Research Reference	SSRN 6734225 — Ondieki Ombachi, 2026
Assessment Type	RAG Corpus Poisoning Security Assessment

Executive Summary

Risk Score	**98/100 — CRITICAL**
Vulnerabilities Found	1
Attack Vectors Tested	3 (QTPI, Document Poisoning, Rank Manipulation)
MHL Detection Rate	100% (0% false quarantine)
Regulatory Frameworks Affected	NIST AI RMF, OWASP LLM Top 10, FATF 2025

This assessment evaluated the target system's resilience against adversarial corpus poisoning attacks — a class of attack first formally documented in SSRN 6734225 (Ondieki Ombachi, 2026). These attacks exploit the retrieval corpus layer of RAG-based AI systems, injecting adversarial documents through legitimate ingestion pathways to collapse system accuracy while generating zero anomalous log signatures.

Attack Results

Attack Vector	β (docs)	Baseline F1	Attacked F1	Degradation	Log Anomalies
QTPI (Prompt Injection)	50	0.919	0.017	**98.2%**	0
Document Poisoning	280	0.919	0.924	+0.5%	0
Rank Manipulation	280	0.919	0.943	+2.6%	0

QTPI achieves 98.4% F1 degradation at the minimum tested injection budget of 50 documents. The agent becomes operationally equivalent to random guessing while producing no anomalous log signatures.

Defense Recommendations

Deploy the Memory Hygiene Layer (MHL) as a pre-ingestion gate in your RAG pipeline. No model retraining required. No infrastructure changes required.

Priority	Action	Effort	Impact
1 — Critical	Deploy CPT (Cryptographic Provenance Tracking)	2 hours	100% QTPI detection
2 — High	Configure FCS threshold at 0.65	1 hour	Reduces FP rate to 0%
3 — High	Enable SRAD sliding window monitoring	1 hour	Corpus-level anomaly detection
4 — Medium	Integrate MHL into document ingestion pipeline	1 day	Full protection
5 — Medium	Set up Prometheus metrics + alerting	2 hours	Operational visibility

Regulatory Mapping

Framework	Gap Identified	MHL Mitigation
NIST AI RMF	No profile for retrieval corpus integrity	First reference implementation
OWASP LLM Top 10	Corpus-level prompt injection not covered	QTPI formalises new LLM01 sub-category
MITRE ATLAS	No technique for corpus manipulation	New attack pattern documented
FINRA Rule 3110	QTPI leaves no audit signature	CPT provides cryptographic audit trail
FinCEN 2024	Agent at F1=0.014 not reasonably designed	CPT + SRAD provide auditable controls
FATF 2025	LLM-generated docs bypass human review	CPT detects source-level anomalies

Generated by CorpusGuard v0.1.0 — github.com/OndiekiFrank/CorpusGuard — Based on SSRN 6734225 — Frankline Ondieki Ombachi — ondiekifrank021@gmail.com